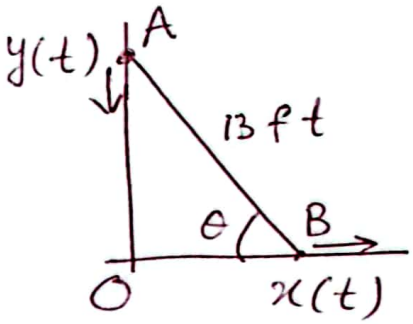
SOLUTION- ASSIGNMENT 3

FALL-2024

MA-110

Q 1



When $x(t) = 12$ ft,

then $\frac{dx(t)}{dt} = 5$ ft/s

(a) In ΔAOB,

$$(\bar{AO})^2 + (\bar{OB})^2 = (\bar{AB})^2$$

$$y^2(t) + x^2(t) = 13^2 = 169$$

$$2y(t)\frac{dy(t)}{dt} + 2x(t)\frac{dx(t)}{dt} = \frac{d}{dt}(169)$$

$$2y(t)\frac{dy(t)}{dt} = 0 - 2x(t)\frac{dx(t)}{dt}$$

When $x(t) = 12$, then $y^2(t) + 12^2 = 13^2$

$$y^2(t) = 169 - 144 = 25 \Rightarrow y(t) = 5 \text{ ft}$$

$$\Rightarrow 2(5)\frac{dy(t)}{dt} = -2(12)(5)$$

$$\Rightarrow \boxed{\frac{dy(t)}{dt} = -12 \text{ ft/s}}$$

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(b) Area of $\Delta AOB = A = \frac{1}{2} x(t)\, y(t)$

$$\frac{dA}{dt} = \frac{1}{2} x(t) \frac{dy(t)}{dt} + \frac{1}{2} y(t) \frac{dx(t)}{dt}$$

$$= \frac{1}{2}(12)(-12) + \frac{1}{2}(5)(5) = \frac{-144}{2} + \frac{25}{2}$$

$$\boxed{\frac{dA}{dt} = -\frac{119}{2} \text{ ft}^2/\text{s}}$$

-ve sign indicates that area is decreasing

(c) $\tan\theta(t) = \frac{y(t)}{x(t)} \Rightarrow \frac{d}{dt}(\tan\theta) = \frac{d}{dt}\left(\frac{y(t)}{x(t)}\right)$

$$\Rightarrow \sec^2\theta \frac{d\theta}{dt} = \frac{-y\frac{dx}{dt} + x\frac{dy}{dt}}{x^2}$$

$$\Rightarrow (1 + \tan^2\theta)\frac{d\theta}{dt} = \frac{x\, dy/dt - y\, dx/dt}{x^2}$$

$$\Rightarrow \left(1 + \frac{y^2}{x^2}\right)\frac{d\theta}{dt} = \frac{x\, dy/dt - y\, dx/dt}{x^2}$$

$$\Rightarrow \frac{x^2 + y^2}{x^2}\frac{d\theta}{dt} = \frac{x\, dy/dt - y\, dx/dt}{x^2}$$

$$\Rightarrow \frac{d\theta}{dt} = \frac{x\, dy/dt - y\, dx/dt}{x^2} \cdot \frac{x^2}{x^2 + y^2}$$

When, $x(t) = 12\,ft$, $\frac{dx}{dt} = 5\,ft/s$,

$y(t) = 5\,ft$, $\frac{dy}{dt} = -12\,ft/s$

Then, $$\frac{d\theta}{dt} = \frac{12(-12) - 5(5)}{12^2 + 5^2} = \frac{-144 - 25}{144 + 25}$$

$$\boxed{\frac{d\theta}{dt} = -1\ rad/s}$$

Q2 $g(x) = x^2\sqrt{5-x}$

$$\frac{dg}{dx} = 2x\sqrt{5-x} + x^2 \frac{1}{2}(5-x)^{-\frac{1}{2}}(-1)$$

$$= 2x\sqrt{5-x} - \frac{x^2}{2\sqrt{5-x}}$$

To find stationary points,

$$\frac{dg}{dx} = 0 \Rightarrow 2x\sqrt{5-x} - \frac{x^2}{2\sqrt{5-x}} = 0$$

$$\Rightarrow 2x\sqrt{5-x} = \frac{x^2}{2\sqrt{5-x}}$$

$$\Rightarrow 4x(5-x) = x^2 \Rightarrow 20x - 4x^2 - x^2 = 0$$

$$5x^2 - 20x = 0 \Rightarrow 5x(x-4) = 0$$

$$\Rightarrow x = 0, \quad x = 4$$

There are two stationary points $x=0$ & $x=4$. To find which is max/min,

$$\frac{d^2g}{dx^2} = \frac{d}{dx}\left(2x\sqrt{5-x} - \frac{x^2}{2\sqrt{5-x}}\right)$$

$$= 2\sqrt{5-x} + \frac{2x(-1)}{2\sqrt{5-x}} - \left\{\frac{2x}{2\sqrt{5-x}} + \frac{x^2}{2}\frac{(5-x)^{-\frac{3}{2}}(-1)}{(-2)}\right\}$$

$$= 2\sqrt{5-x} - \frac{x}{\sqrt{5-x}} - \frac{x}{\sqrt{5-x}} - \frac{x^2}{2^2(5-x)^{3/2}}$$

$$= 2\sqrt{5-x} + \frac{x}{2\sqrt{5-x}}\left[\frac{x}{4(5-x)} - 2\right]$$

At $x=0$,

$$\left.\frac{d^2g}{dx^2}\right|_{x=0} = 2\sqrt{5} + 0 = 2\sqrt{5} > 0$$

i.e. $x=0$ is a point of minimum.

At $x=4$,

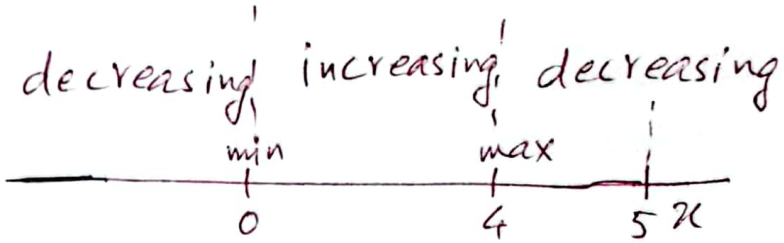
$$\left.\frac{d^2g}{dx^2}\right|_{x=4} = 2\sqrt{5-4} + \frac{4}{\sqrt{5-4}}\left[\frac{4}{4(5-4)} - 2\right]$$

$$= 2 + 4[1-2] = 2 - 4 = -2 < 0$$

i.e. $x=4$ is a point of max.

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Function $g(x)$ is increasing in the interval $(0,4)$, decreasing in $(-\infty,0)$ and decreasing in $(4,5)$.

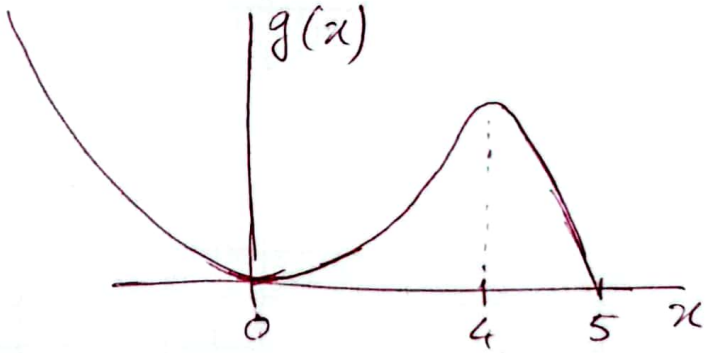


(b) Function's extreme values:

min.: $g(0) = 0^2\sqrt{5-0} = 0$

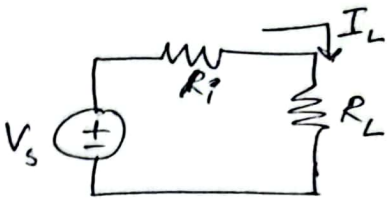
max.: $g(4) = 4^2\sqrt{5-4} = 16$

(c)



The function will not go beyond $x = 5$. For $x > 5$, $(5-x)$ will be negative, and square root of negative numbers does not exist in real domain.

Q3



Power dissipated at R_L is

$$P = \frac{V_s^2 R_L}{(R_i + R_L)^2}$$

$$\frac{dP}{dR_L} = V_s^2 \left[\frac{1}{(R_i + R_L)^2} + R_L (-2)(R_i + R_L)^{-3} \right]$$

$$= V_s^2 \left[\frac{1}{(R_i + R_L)^2} - \frac{2R_L}{(R_i + R_L)^3} \right]$$

$$= \frac{V_s^2}{(R_i + R_L)^2} \left[1 - \frac{2R_L}{R_i + R_L} \right]$$

To find the point of maximum,

$$\frac{dP}{dR_L} = 0 \Rightarrow \frac{V_s^2}{(R_i + R_L)^2} \left[\frac{R_i + R_L - 2R_L}{R_i + R_L} \right] = 0$$

$$\frac{R_i - R_L}{R_i + R_L} = 0 \Rightarrow \boxed{R_L = R_i}$$

To determine, whether $R_L = R_i$ is a point of max. or min. we take 2nd derivative. (2C)

$$\frac{d^2P}{dR_L^2} = \frac{d}{dR_L} \left[\frac{V_s^2 (R_i - R_L)}{(R_i + R_L)^3} \right] = \left[\frac{-3(R_i - R_L)}{(R_i + R_L)^4} - \frac{1}{(R_i + R_L)^3} \right] V_s^2$$

At $R_L = R_i$,

$$\left. \frac{d^2P}{dR_L^2} \right|_{R_L = R_i} = V_s^2 \left[0 - \frac{1}{(R_i + R_i)^3} \right] < 0$$

i.e. at $R_L = R_i$, power is maximum.

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Q4 (a) At $t = 0, 3.2, 5.1$ seconds, $v = 0$ as obvious from the graph, i.e. the particle is not moving at $t = 0, 3.2, 5.1$ seconds.

(b) At $t = 1.9$ sec, 4 sec. and 6 sec., the tangents to the velocity-time graph are horizontal lines, i.e. $\left.\frac{dv(t)}{dt}\right|_{t=1.9,4,6} = 0$

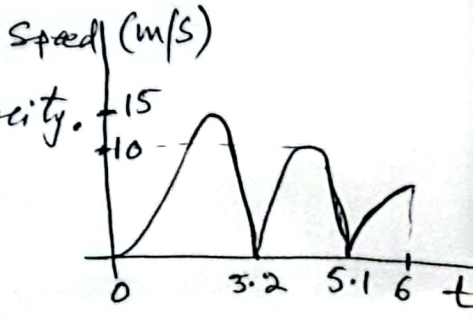
i.e. acceleration is zero at $t = 1.9, 4, 6$ seconds.

(c)(i) Acceleration means increase in velocity. As obvious from the graph, the velocity is increasing in intervals (0, 1.9 sec.), (5.1 sec., 6.1 sec.). Moreover, the velocity is also increasing in (3.2, 4) sec. but in opposite direction.

(ii) The particle is decelerating in (1.9, 3.2 sec.) and (4, 5.1 sec.)

(d) The particle changes direction at $t = 3.2$ sec. and $t = 5.1$ sec.

(e)(i) speed is the magnitude of velocity.



(ii)

